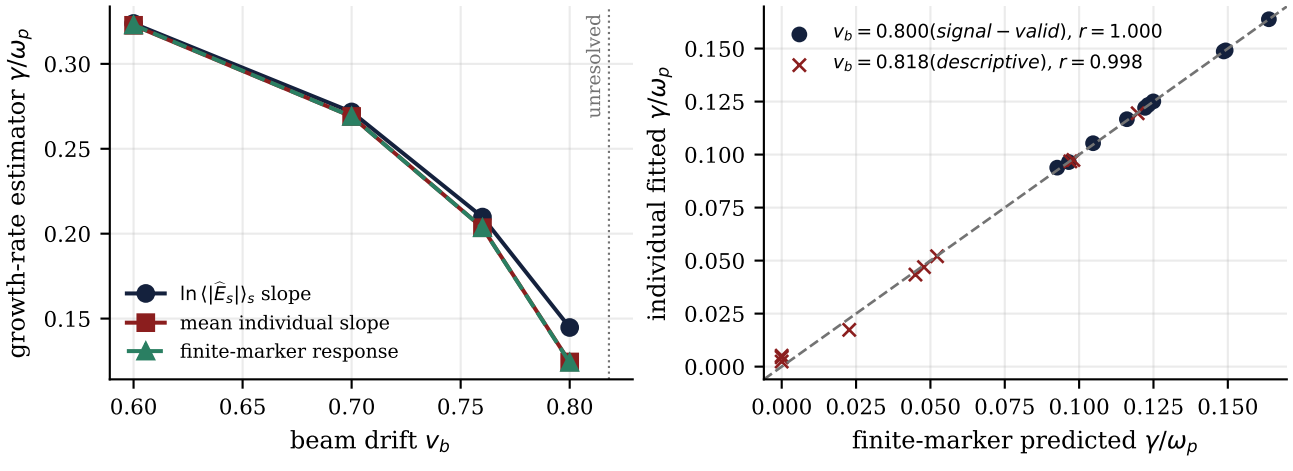
(a) Estimator dependence
growth-rate estimator $\gamma/\omega_p$
0.30
0.25
0.20
0.15
unresolved
$\ln\langle|\widehat{E}_s|\rangle_s$ slope
mean individual slope
finite-marker response
0.60
0.65
0.70
0.75
0.80
beam drift $v_b$
(b) Loading explains seed spread
individual fitted $\gamma/\omega_p$
$v_b = 0.800$(*signal − valid*), $r = 1.000$
$v_b = 0.818$(*descriptive*), $r = 0.998$
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.000
0.025
0.050
0.075
0.100
0.125
0.150
finite-marker predicted $\gamma/\omega_p$